

Current Mirror

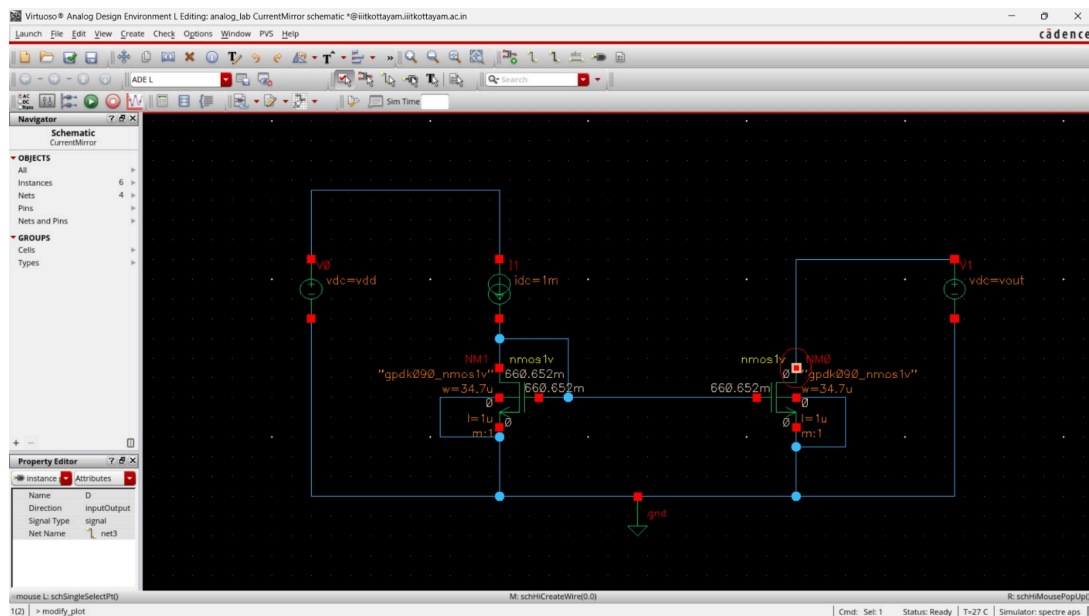
Github repo link:

https://github.com/Abhiram-Jayakumar-07/AnalogElectronicsLab_2025BEC0020/tree/main/Current%20Mirror

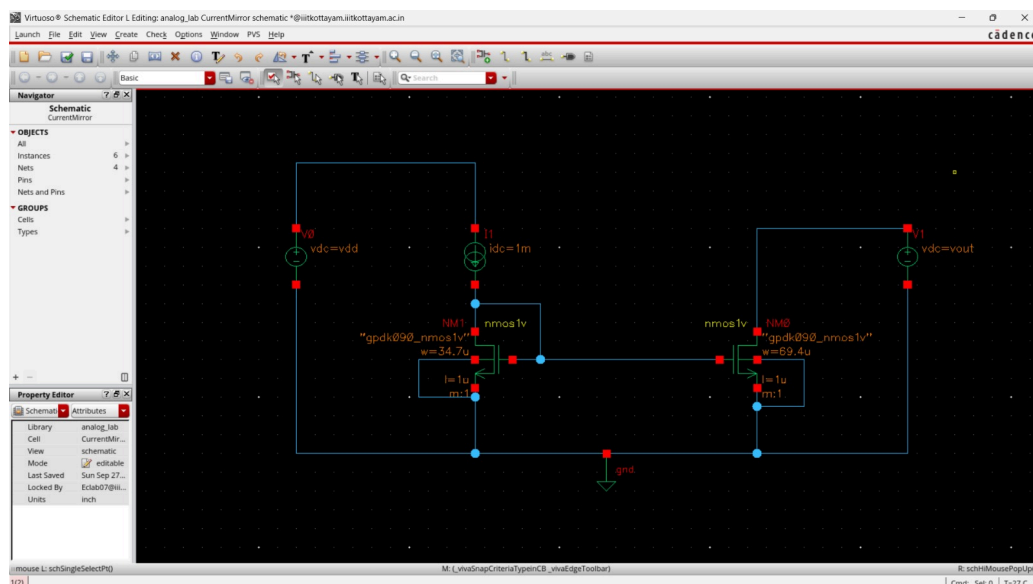
Simple current mirror:

Schematic:

1 mA:

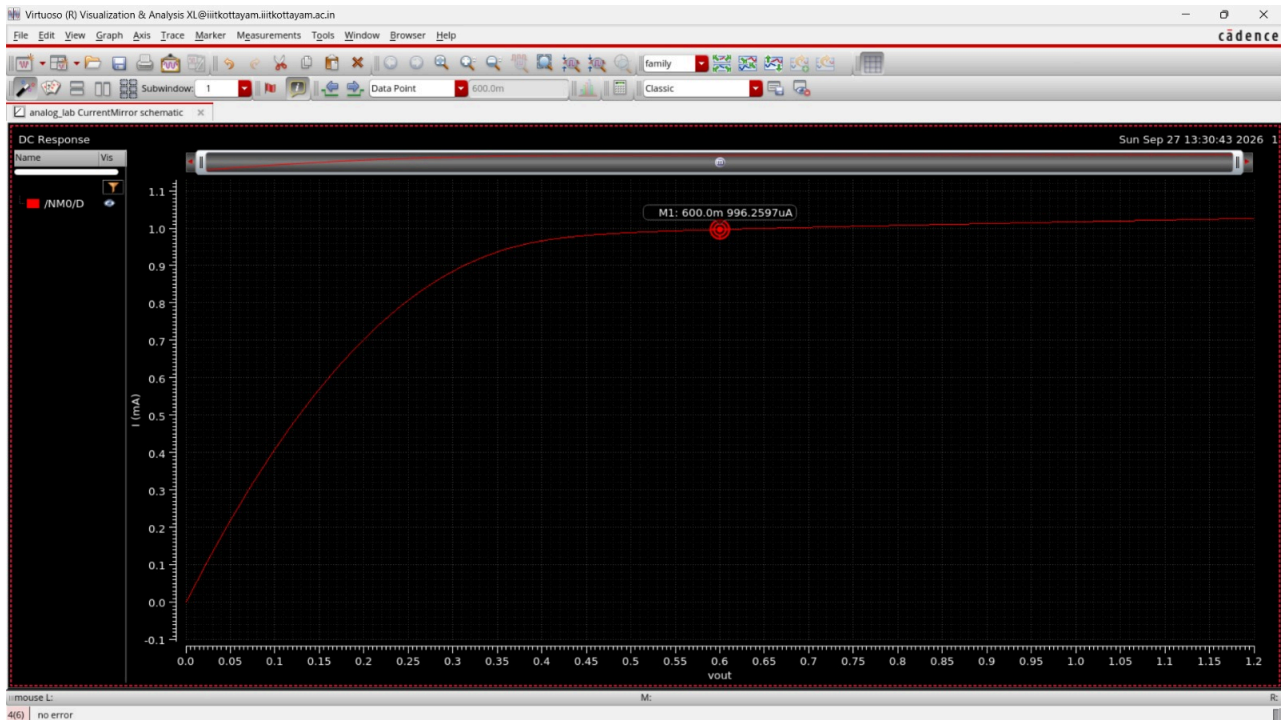


2 mA:

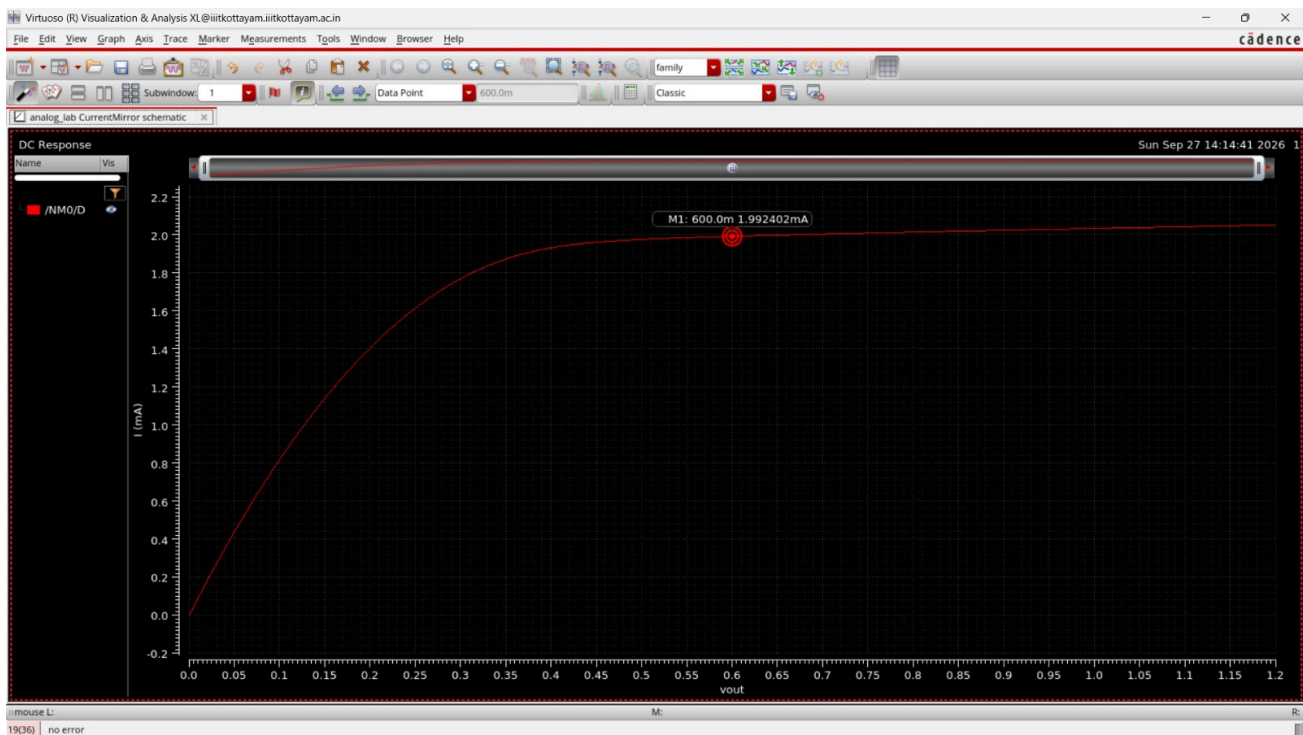


Results:

1 mA:



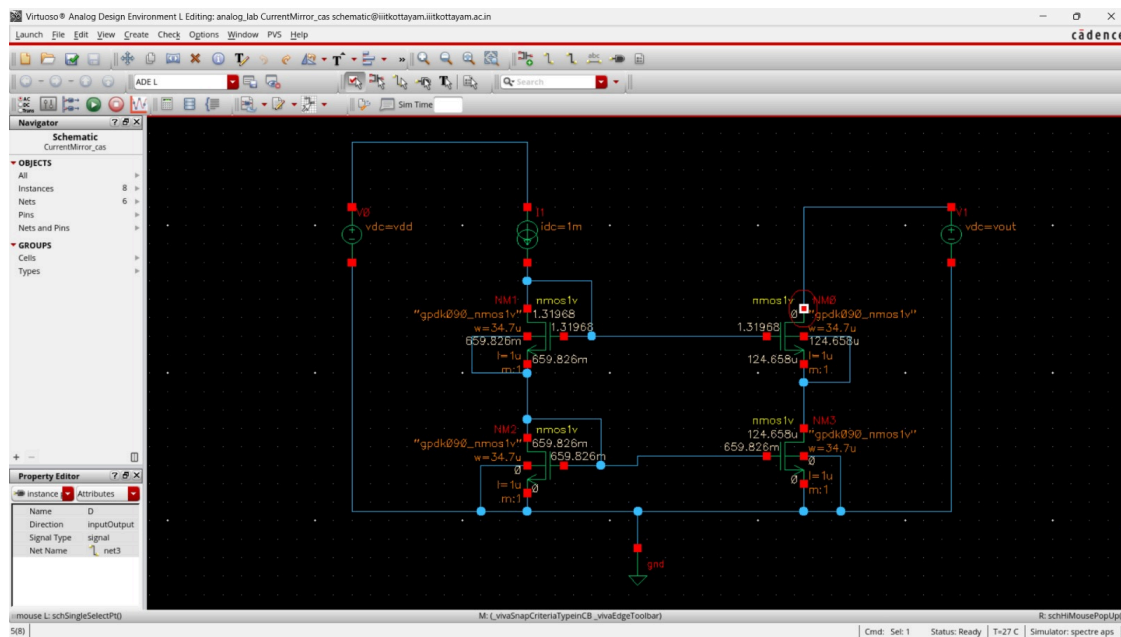
2 mA:



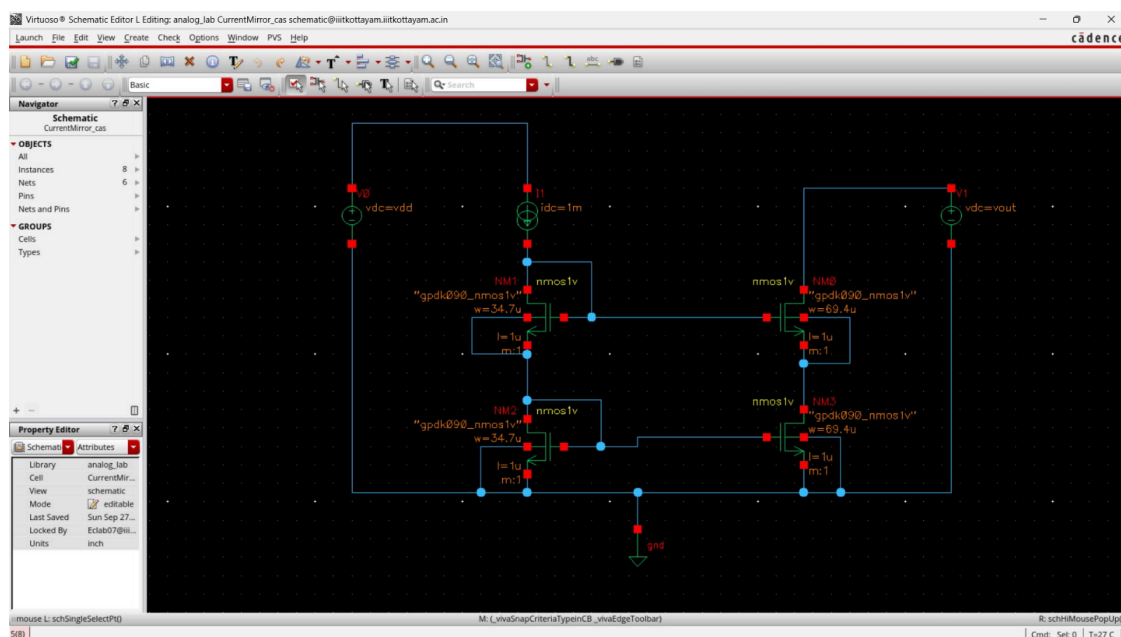
Cascode current mirror:

Schematic:

1 mA:

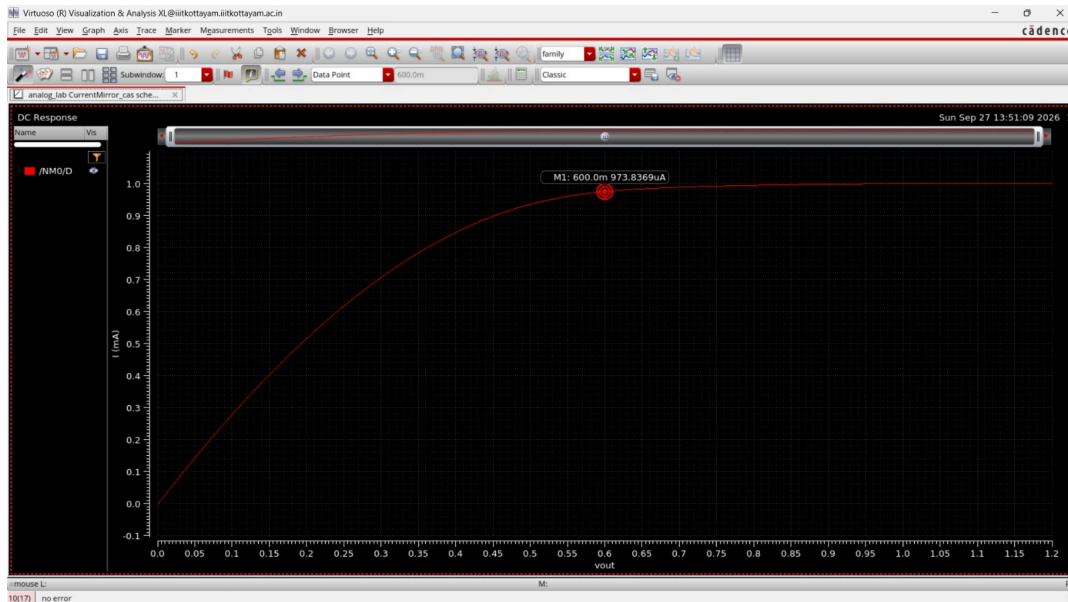


2 mA:

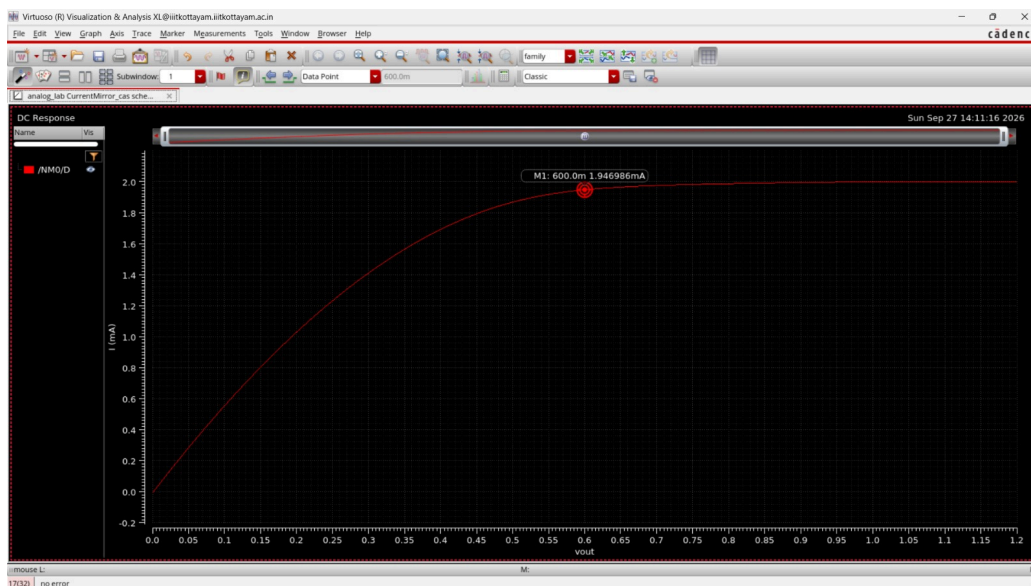


Results:

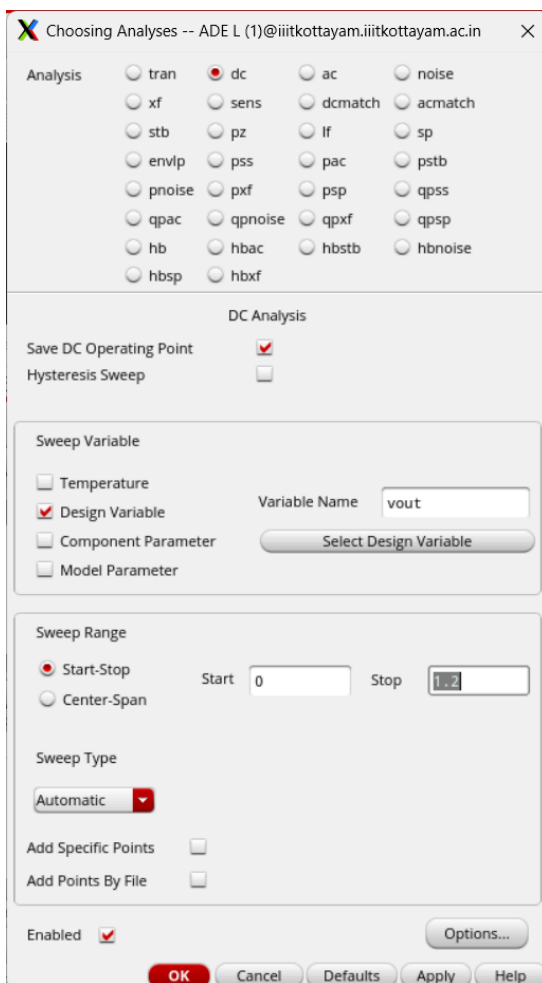
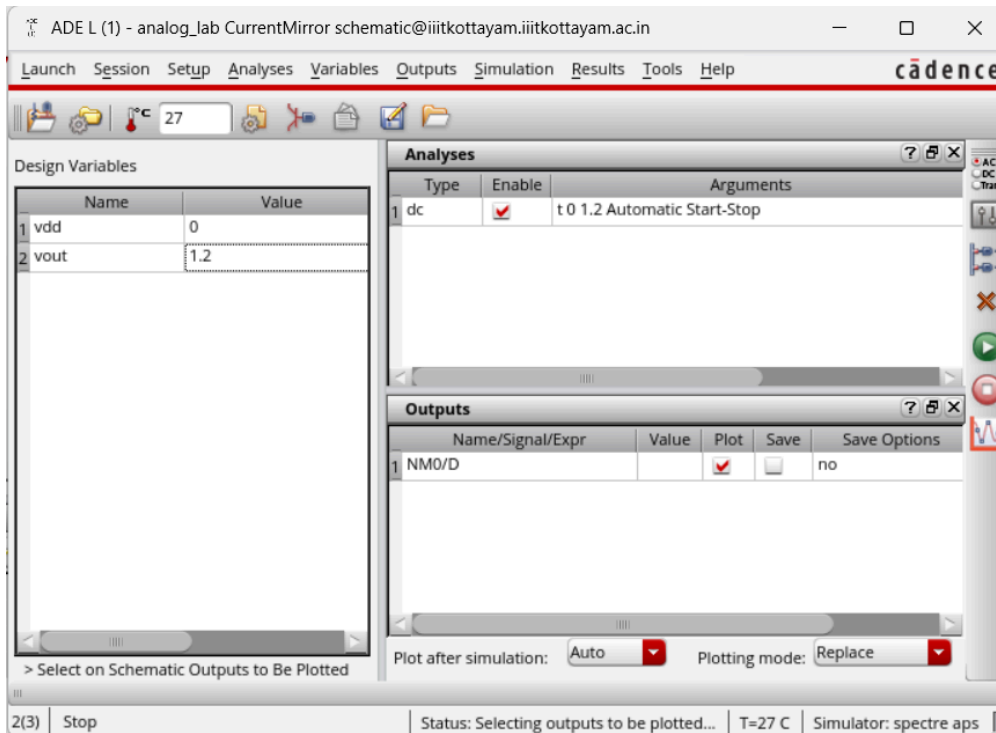
1 mA:



2 mA:



Simulation:



Calculations:

Derived from the $g_m/I_D = 15 \text{ V}^{-1}$ characterization curve:

Simple Current Mirror

- 1 mA Output (1:1 Mirror Ratio):
 - $I_{ref} = 1 \text{ mA}$
 - Reference Device (M1): $W = 1 / 288 = 34.7 \text{ } \mu\text{m}$, $nf=2$
 - Mirror Device (M2): $W = 34.7 \text{ } \mu\text{m}$, $nf=2$
- 2 mA Output (1:2 Mirror Ratio):
 - $I_{ref} = 1 \text{ mA}$
 - Reference Device (M1): $W = 34.7 \text{ } \mu\text{m}$, $nf=2$
 - Mirror Device (M2): $W = 69.4 \text{ } \mu\text{m}$, $nf=4$

Cascode Current Mirror

- 1 mA Output (1:1 Mirror Ratio):
 - $I_{ref} = 1 \text{ mA}$
 - Reference Leg (NM1, NM2): $W = 34.7 \text{ } \mu\text{m}$, $nf=2$
 - Output Leg (NM0, NM3): $W = 34.7 \text{ } \mu\text{m}$, $nf=2$
- 2 mA Output (1:2 Mirror Ratio):
 - $I_{ref} = 1 \text{ mA}$
 - Reference Leg (NM1, NM2): $W = 34.7 \text{ } \mu\text{m}$, $nf=2$
 - Output Leg (NM0, NM3): $W = 69.4 \text{ } \mu\text{m}$, $nf=4$

Inference:

The cascode current mirror is the superior current source because the addition of the cascode transistor increases the small-signal output resistance by a factor of approximately $g_m r_o$. This high output impedance suppresses channel-length modulation effects, providing a much flatter I_{out} vs V_{out} characteristic curve and ensuring precise current mirroring compared to a simple current mirror.